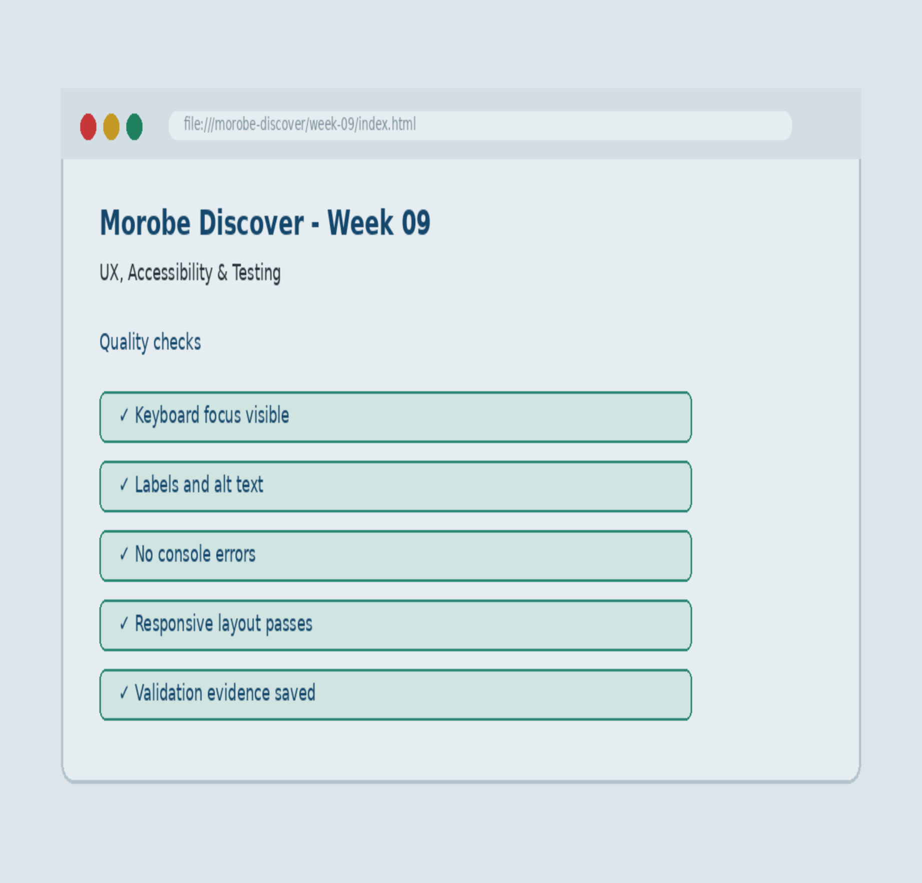
IS229 WEB DESIGN

Week 09
UX, Accessibility & Testing

Ensure the site is usable for people with different abilities and optimized for performance.



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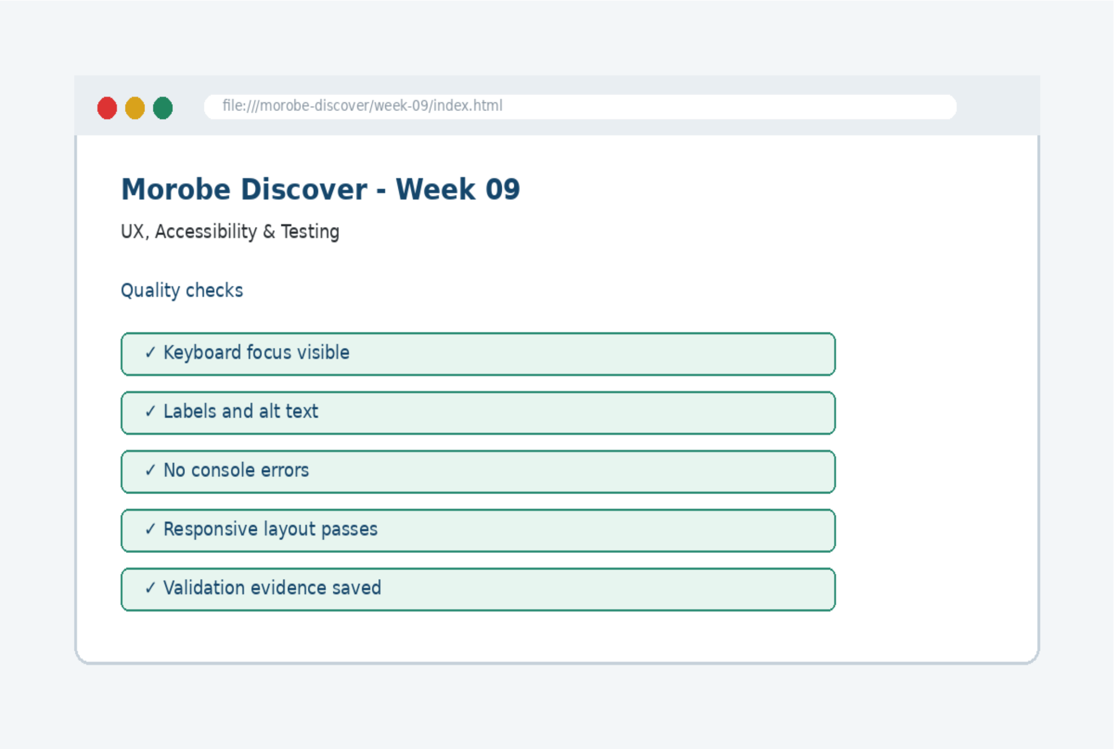
Learning outcomes

Ensure the site is usable for people with different abilities and optimized for performance.
Explain the core programming/design concept for this week
Read and modify key code patterns
Test the weekly project increment
Prepare evidence for GitHub submission

Today's problem



Why it matters

Accessibility is not an optional add-on; it is part of professional interface quality.

Project before -> after

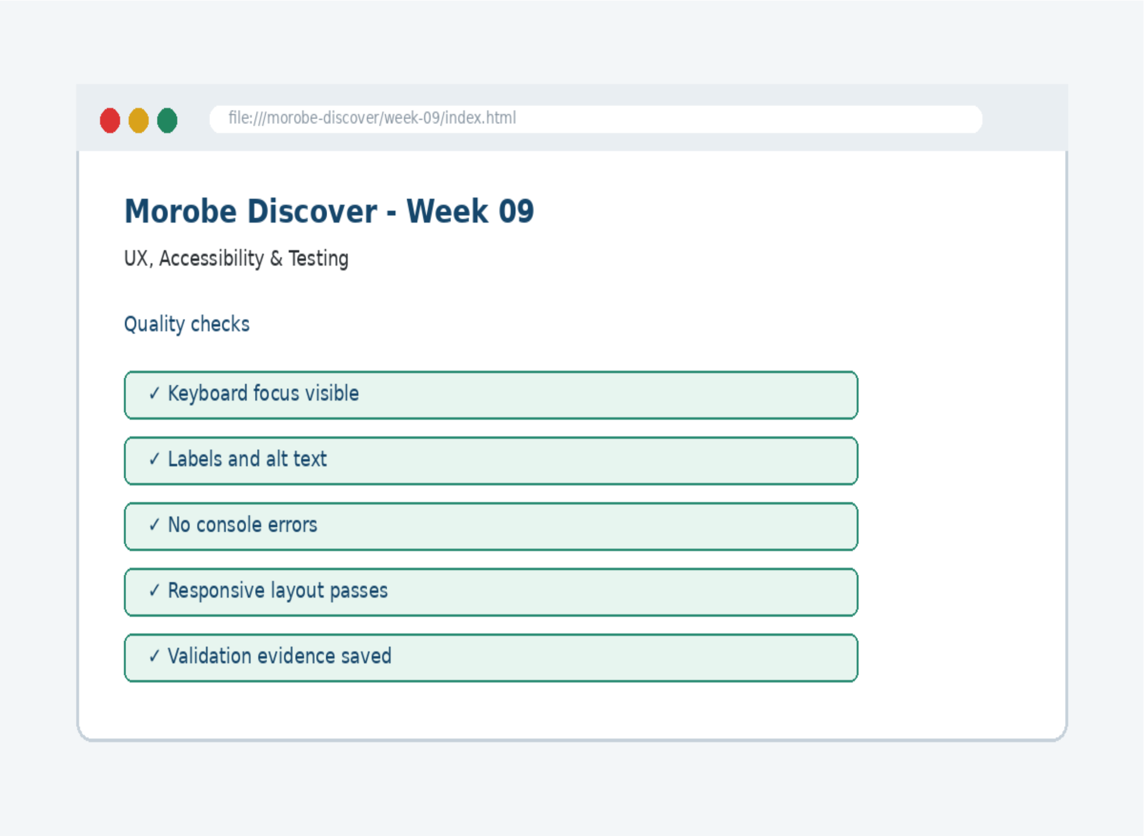
Before Week 09

Week 08 created dynamic data rendering and filtering.

After Week 09

The site is tested as a user experience, including keyboard access, labels, focus and feedback.

What we build this week



Add or confirm labels and alt text
Add visible focus styles
Test the journey using keyboard only
Document validation and accessibility checks
Fix the highest-impact issues first

Core principle

Accessibility is not an optional add-on; it is part of professional interface quality.

Keep in mind

A good web developer can point to the code, the browser result, and the user reason behind the decision.

Key concepts

Concept 1

Native HTML gives built-in accessibility

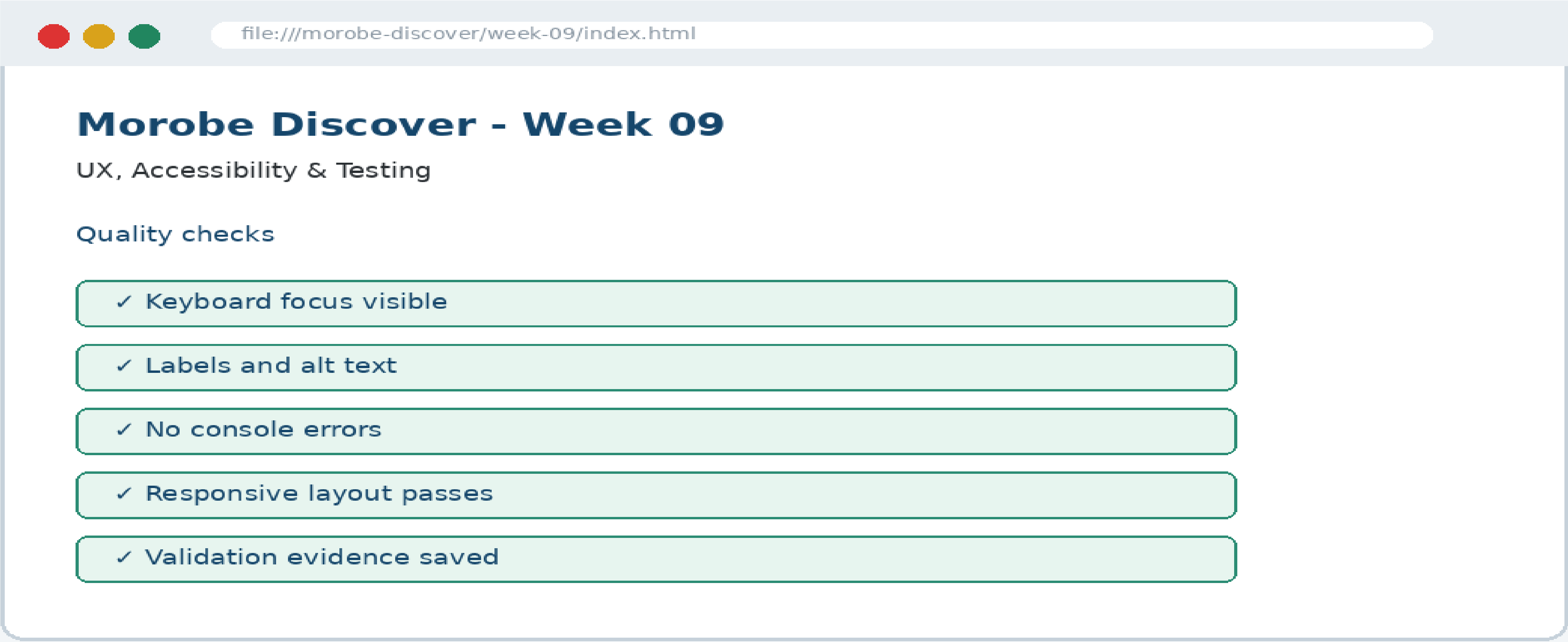
Concept 2

Visible focus shows keyboard position

Concept 3

Testing must include both successful and failure paths

Project anatomy



Key code pattern A

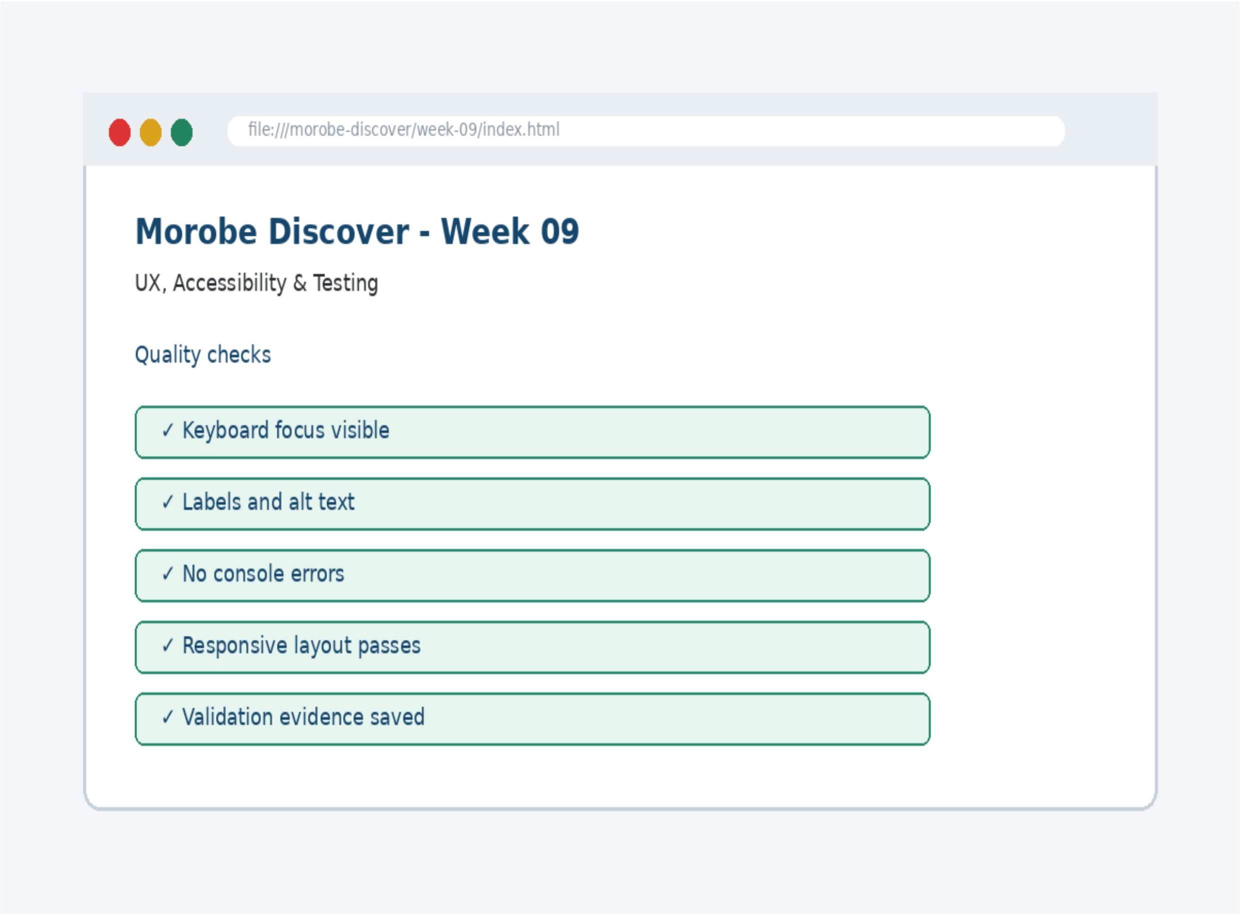
```
.skip-link:focus {
  position: static;
  padding: .75rem;
}
```

Explain it

Which file contains this code?
Which line creates the visible change?
What should appear in the browser?

Code -> visual result

```css
.skip-link:focus {
  position: static;
  padding: .75rem;
}
```



What changed?

Project file changed for UX, Accessibility & Testing
The browser output is now closer to the required weekly evidence
The change supports a user task or quality requirement
The change can be tested and committed

Key code pattern B

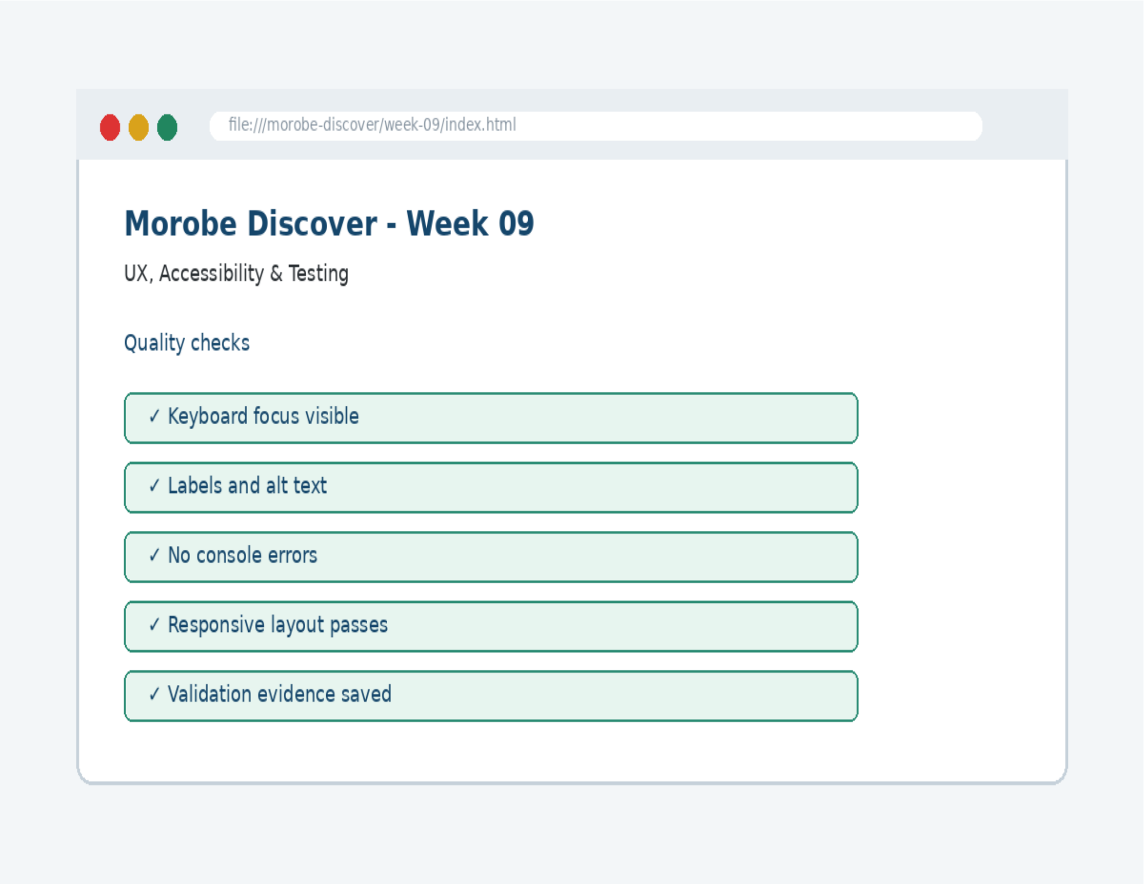
```
<label for="destinationSearch">Search
destinations</label>
<input id="destinationSearch" type="search">
```

Why this pattern is needed

The second pattern usually applies the same idea to a different component, state, layout or data flow.

Apply it to Morobe Discover

Add or confirm labels and alt text
Add visible focus styles
Test the journey using keyboard only
Document validation and accessibility checks



Compare the approaches

Weak approach

Memorise the definition
Copy code without understanding
No browser evidence
No test note

Professional approach

Explain the concept
Apply the correct code pattern
Inspect the result
Commit and describe the evidence

Common mistake

Mistake

The page appears unchanged, but the student assumes the feature is complete.

Correction

Check the correct file, refresh the browser, inspect DevTools, and verify the expected output.

Why does this fail?

```
// Broken checkpoint
// File changed, but the browser result does not
change.
// Possible causes:
// - wrong file
// - file not linked
// - syntax error
// - browser not refreshed
```

Class question

Which cause would you check first, and why?

Fix the problem

Confirm the browser is opening the correct file
Check links to CSS/JS/data files
Inspect the console for errors
Use a small working example
Test again before committing

Class activity

Activity - 7 minutes

Find three barriers that might not be obvious when using only a mouse.

Work in pairs
Point to the project evidence
Prepare one answer to share

Implementation steps 1-3

Add or confirm labels and alt text
Add visible focus styles
Test the journey using keyboard only

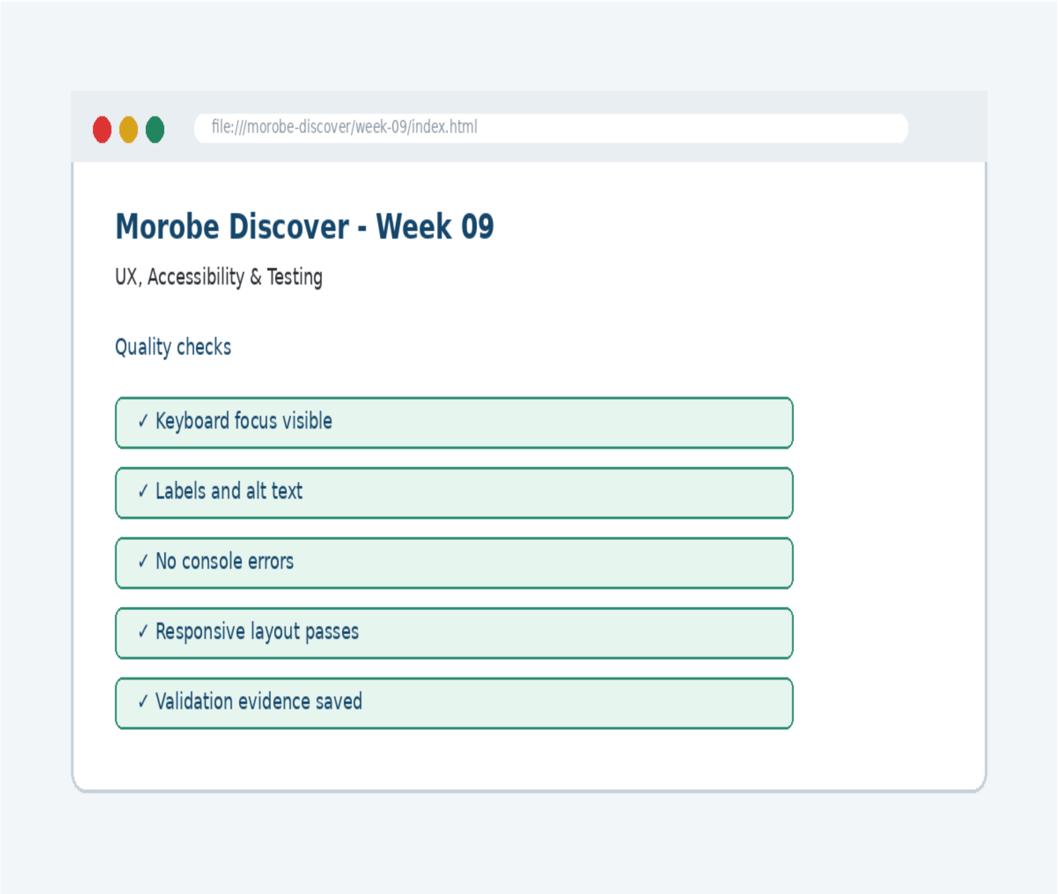
Implementation steps 4-5

Document validation and accessibility checks
Fix the highest-impact issues first

LIVE DEMO

Demo checklist

1. Open the current project
2. Locate the affected file
3. Add or modify the key code
4. Refresh the browser
5. Inspect the result
6. Explain the change



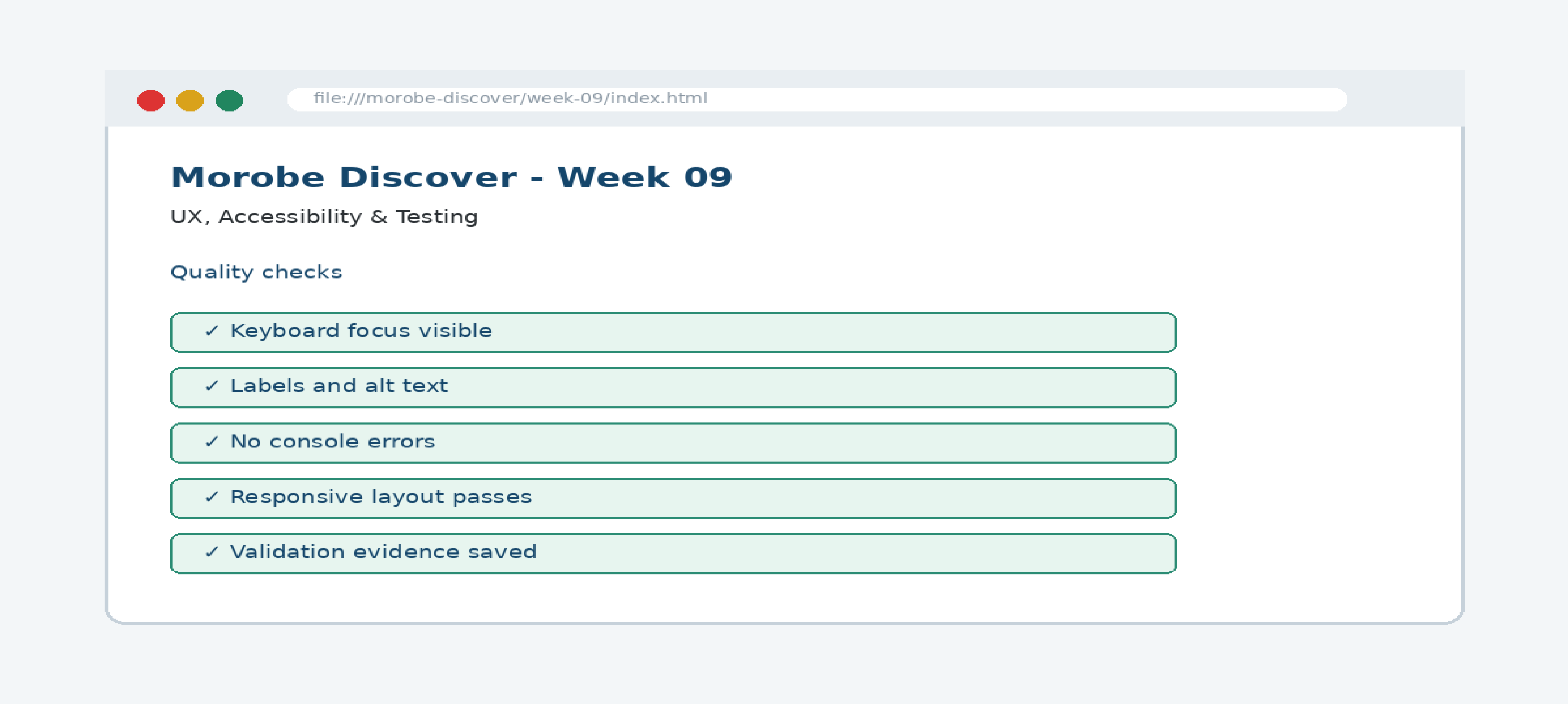
Predict the output

```
<label for="destinationSearch">Search
destinations</label>
<input id="destinationSearch" type="search">
```

Prediction prompt

What should change on screen when this code is applied correctly?

Browser result



Testing the feature

Tab through the page
Run HTML/CSS validation
Use Lighthouse or manual checklist
Confirm errors are visible and understandable

Inspect with browser tools

Inspection path

Right click -> Inspect
Elements panel -> structure/classes
Console -> JavaScript errors
Network/Application -> data/storage when relevant

Evidence to save

Screenshot
Short test note
Changed files
Git commit message

Git evidence

```
git add .
git commit -m "Week 09 UX, Accessibility & Testing
project increment"
```

Why it matters

Git history shows progression from Week 01 to final deployment. It is part of the assessment evidence.

Complete source code

```
Repository:
https://github.com/YOUR-USERNAME/is229-morobe-discover

Weekly folder:
/week-09/

Complete project:
/complete-project/
```

Reminder

Do not paste a whole file into a slide. Use the repository for complete code.

Mini challenge

Challenge

Modify one part of the Week 09 implementation and explain what changed in the browser.

State the file edited
State the line/section changed
Show the before and after result

What counts as evidence?

Technical evidence

Changed code files
Browser screenshot
Test note
Git commit

Understanding evidence

Code explanation
Why this method fits
Connection to previous week
Connection to next week

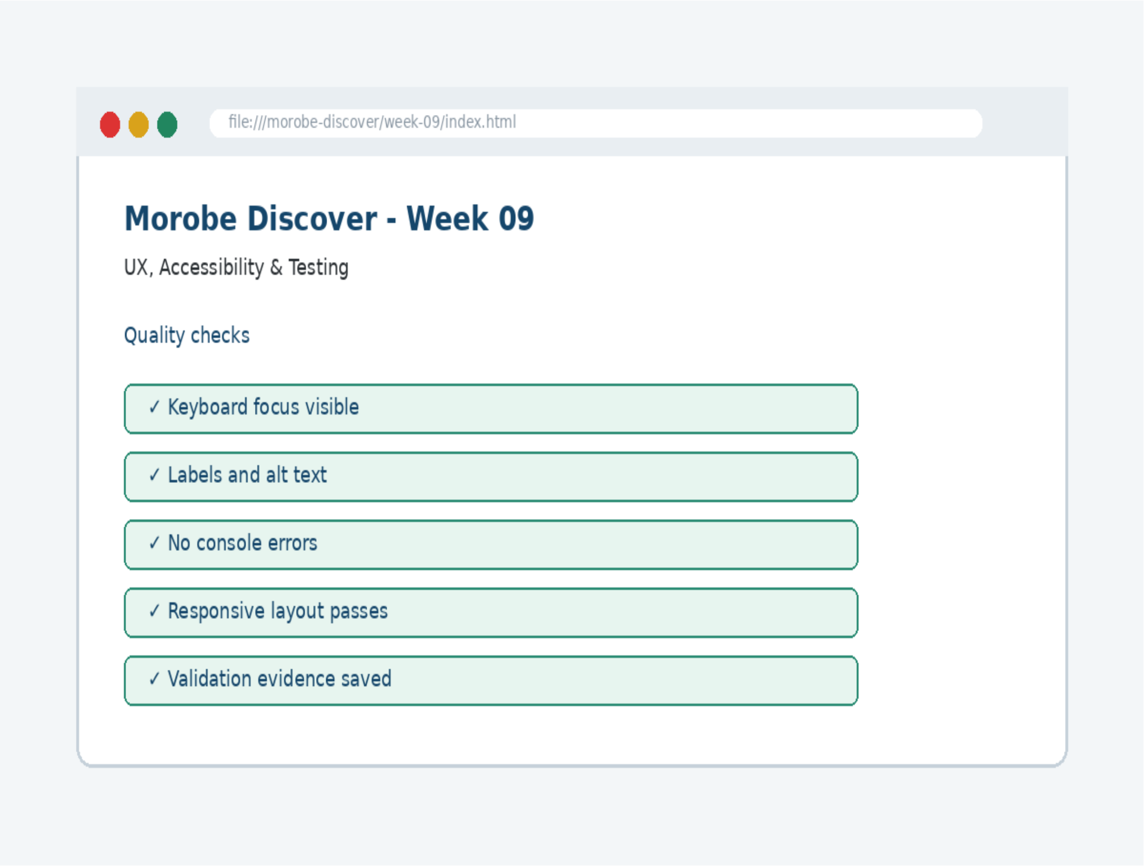
Project lab handoff

- Add or confirm labels and alt text
- Add visible focus styles
- Test the journey using keyboard only
- Document validation and accessibility checks
- Fix the highest-impact issues first

Definition of done

Week 09 feature is implemented
Project still preserves earlier week features
Browser result matches the expected output
Test note and screenshot are saved
Git commit message is meaningful

What we built today



The site is tested as a user experience, including keyboard access, labels, focus and feedback.
Evidence: Alt text, labels, visible keyboard focus styles and interactive state checks
Technology: HTML / CSS / Testing Tools

Lecture summary

Accessibility is not an optional add-on; it is part of professional interface quality.
Native HTML gives built-in accessibility
Visible focus shows keyboard position
Testing and Git are part of development, not extras

Prepare for next class

Finish this week’s lab before the next class
Read the next topic in advance
Bring the current project ready to extend
Be ready to explain one code decision

Exit question

How did Week 09 improve Morobe Discover, and what will the next week build on?